Constraints on Combining Pole Certificates

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Abstract

Two elementary obstructions distinguish a joint pole certificate from two independent certificates. First, a common nondegenerate conic parameter forces the two inputs to have the same rational squareclass. Two explicit prepared inputs admit no joint certificate with a common parameter, including the degenerate values. Second, reflection-symmetric contact sets admit an unbounded family on which the product of two scaled contact functions is a square. We also identify the prime divisors responsible for overlap between two scaled contact sets. These statements concern the specified combinations; they are not lower bounds for arbitrary Diophantine definitions.

1 A common parameter in the pole conics

Call a rational number b *prepared* if its reduced denominator is a square and

$$0 < b < 17, \qquad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2},$$
$$v_2(b) \geq 3, \qquad v_3(b) \geq 1, \qquad v_{17}(b) \geq 2.$$

For $B = 17b$ and $V = B(B - b^2)$, the pole equations are

$$p^2 = B(1 - e)(1 + 3e), \qquad Bq^2 = (s^2 - eV)(s^2 + 3eV). \tag{1}$$

All witnesses in this section are rational; zero witnesses are allowed. The independent realization of these equations on prepared inputs is the arithmetic input stated in [1].

Lemma 1.1. *Let $b_0, b_1 \in \mathbb{Q}^\times$. If both pole systems have a solution with the same parameter $e \notin \{1, -1/3\}$, then b_0/b_1 is a rational square.*

Proof. Put $B_i = 17b_i$. The first equations give $p_i^2 = B_i(1 - e)(1 + 3e) \neq 0$. Their quotient is $b_0/b_1 = (p_0/p_1)^2$. □

Theorem 1.2. *The prepared numbers*

$$b_0 = \left(\frac{204}{55}\right)^2, \qquad b_1 = 409\left(\frac{204}{1001}\right)^2$$

have no pair of pole certificates with a common value of e.

Proof. Both reduced denominators are squares and both numbers lie in $(0, 17)$. Their valuations at $2, 3, 17$ are respectively $4, 2, 2$. The additional factor 409 is 1 modulo 8, 3, and 17, so it is a square in all three local fields. Thus both numbers are prepared. But

$$\frac{b_1}{b_0} = 409\left(\frac{55}{1001}\right)^2$$

is not a rational square. Lemma 1.1 therefore forces $e = 1$ or $e = -1/3$ in any proposed joint certificate. For b_0, neither value permits the second pole equation: this is the exact rank-certified obstruction proved in [2, Theorem 1.2 and Sections 2–3]. That obstruction permits zero values of q and hence applies to the present convention. Both remaining values of e are excluded. □

This theorem does not identify the parameters of the rational conic charts. Indeed

$$e_i = \frac{\tau^2 - B_i}{\tau^2 + 3B_i}$$

usually gives different e_i for the same τ. Thus a joint realization with a common chart parameter τ is a separate arithmetic question. Zariski density of the two independent witness sets, proved in [3], does not assert that their product meets the subvariety on which two parameters are equal.

2 A square product from symmetric contacts

For $r \in \mathbb{Q}^\times$, let

$$\mathrm{Odd}(r) = \prod_{p:\, v_p(r) \text{ odd}} p.$$

The product includes both positive and negative odd valuations; it is finite. A nonzero rational square has $\mathrm{Odd} = 1$.

Consider the eight contacts

$$\Theta = \{-3, -2, -1, 0, 2, 3, 4, 5\} = \{1 \pm 1, 1 \pm 2, 1 \pm 3, 1 \pm 4\}.$$

Define

$$F(t) = \prod_{j=1}^{4} ((t-1)^2 - j^2), \qquad R(t) = \frac{F(t)}{(t-1)^{10}}.$$

Both functions are invariant under $t \mapsto 2 - t$.

Theorem 2.1. *For every rational $a > 1$, put*

$$\lambda = \frac{2}{a+1}, \qquad t_0 = \lambda, \qquad t_1 = a\lambda.$$

Both $R(t_i)$ are defined, nonzero, and positive, and

$$R(t_0) = R(t_1), \qquad \mathrm{Odd}(R(t_0)R(t_1)) = 1.$$

Consequently no constants C, C_0 can make

$$h(a) \le C \log \mathrm{Odd}(R(\lambda)R(a\lambda)) + C_0$$

hold for all admissible positive rational a, λ. The conclusion remains true if a is restricted to $a = \beta^4 + T^4$ for any one fixed rational T with $|T| > 1$ and arbitrary rational β.

Proof. The identity $t_0 + t_1 = 2$ gives $R(t_0) = R(t_1)$. Since $0 < t_0 < 1 < t_1 < 2$, neither argument is a contact or the pole 1. At either argument $(t_i - 1)^2 < 1$. All four factors of $F(t_i)$ are therefore negative, so their product is positive; the denominator is positive. The product of the two equal values is a nonzero rational square.

Letting a range through positive integers tending to infinity contradicts the proposed height inequality. For the stated restriction, fix T and let β range through positive integers. Then $a > 1$ and $h(a) \ge \log a$ tends to infinity, whereas the odd radical remains one. □

More generally, if a rational contact function satisfies $R(2c - t) = R(t)$ with $c \neq 0$, the substitution $\lambda = 2c/(a + 1)$ makes the two values equal wherever both are defined. An unbounded admissible set of a then gives the same obstruction.

Theorem 2.1 concerns the product $R(\lambda)R(a\lambda)$. Multiplication or division by a nonzero rational square does not alter its conclusion. Thus the odd-valuation argument supplies no information on this family if that product is used as the squareclass of one pole input. The theorem does not assert that all the other equations of a proposed definition are soluble on that family. It also does not apply to the two independent pole certificates.

3 Where common contact primes occur

Let $\Theta \subset \mathbb{Q}$ be finite. Choose a finite set S of primes containing the prime divisors of the nonzero contact numerators, their denominators, and the nonzero differences of two contacts. For $t \in \mathbb{Q} \setminus \Theta$, let $M_\Theta(t)$ be the product of primes $p \notin S$ such that $v_p(t - \theta) > 0$ is odd for some $\theta \in \Theta$.

Proposition 3.1. *Let $a, \lambda \in \mathbb{Q}^\times$, let $p \notin S$, and suppose that $v_p(a) = 0$. If p belongs to both $M_\Theta(\lambda)$ and $M_\Theta(a\lambda)$, then precisely one of the following occurs:*

1. *both contacts are zero, and $0 \in \Theta$;*
2. *the contacts are nonzero $\theta, \eta \in \Theta$, and $v_p(a\theta - \eta) > 0$.*

In the first case the two contact valuations are equal.

Proof. At each of the two arguments there is exactly one contact modulo p, by the choice of S. Write them as θ, η. Then $\lambda \equiv \theta$ and $a\lambda \equiv \eta \pmod{p}$. If $\theta = 0$, the unit condition on a forces $\eta = 0$; the converse follows in the same way. Their valuations agree because a is a unit. Otherwise both contacts are units, and subtracting gives $a\theta - \eta \equiv 0 \pmod{p}$. □

In particular, a zero contact shared by the two scales disappears from the odd radical of the product whenever a is a square: the corresponding factors multiply to $a\lambda^2$. The other possible overlaps are governed by the fixed ratio set

$$\mathcal{R} = \{\eta/\theta : \theta, \eta \in \Theta \setminus \{0\}\}.$$

If $a \notin \mathcal{R}$ and $a = A/B$ is reduced, every prime in the second case divides the numerator of $a - r$ for some $r \in \mathcal{R}$. Consequently their product $C(a)$ satisfies

$$\log C(a) \leq |\mathcal{R}|h(a) + C_\Theta.$$

Indeed, write $r = u_r/v_r$ in lowest terms. Each relevant prime divides $v_rA - u_rB$, whose absolute value is at most $(|v_r| + |u_r|)\exp(h(a))$, and multiply these bounds. This estimate explains the additional height term in a direct overlap argument. It does not bound that term by a constant independent of a, nor does it exclude a sharper argument using further arithmetic information.

Exact verification

The accompanying program in `computation/pole_parameters`,

`check_pole_merger_obstructions.py`,

checks the prepared rational inputs, their nonsquare ratio, the reflection identity coefficient by coefficient, and the scaled rational-function identity. Its certificate records the precise scope of these checks. The elliptic rank certificate used in Theorem 1.2 remains the independently reproducible certificate accompanying [2].

References

[1] *Arithmetic dependencies of the construction with 29 variables*, companion paper, 8 November 2026, `papers/arithmetic_dependencies`.

[2] *Obstructions to fixed parameters in pole certificates*, companion paper, 9 November 2026, `fixed_pole_parameters_20261109_v16.tex`.

[3] *Zariski density of the prepared pole witnesses*, companion paper, 9 November 2026, `pole_witness_density_20261109_v14.tex`.